

EXECUTIVE PROFILE

- Core Identity & Experience:** Gameplay & Systems Engineer with 3.5+ years of professional studio experience developing high-performance mobile titles in Unity (C#), with 3 live commercial games published on Google Play Store.
- Engineered Subsystems:** Production specialist across modular game subsystems—including procedural puzzle generators, native Android/iOS bridge pipelines, deterministic logic solvers, and non-blocking OTA level sync maintaining stable 60 FPS.
- AI-Augmented Engineering (Claude & ChatGPT/Codex):** Advanced workflow proficiency utilizing Claude 3.5/3.7, ChatGPT, and OpenAI Codex for accelerated systems architecture, algorithmic verification, rapid shader prototyping, and automated editor tooling.
- Forward Vision & Growth:** Committed to architecting scalable, data-driven game loops, mastering next-gen multiplayer frameworks, and driving technical gameplay excellence in collaborative game studios.

PRODUCTION TRACK RECORD (KPIS)

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| <div>3.5+ Years STUDIO EXPERIENCE</div> <div><div><div>Vasundhara Game Studio, Surat, GJ (Dec 2025 – Current)</div><div>StudioKrew, Delhi (Remote) (Mar 2024 – May 2024)</div><div>Appzia, Pune (June 2024 – July 2025)</div></div></div>                                                      | <div>3 Shipped Titles GOOGLE PLAY COMMERCIAL</div> <div><div><div>Calm Jigsaw <a href="#">[Store ↗]</a></div><div>MeowSweeper <a href="#">[Store ↗]</a></div><div>Calm Solitaire <a href="#">[Store ↗]</a></div></div></div>                                                                                                                                                |
| <div>Engineered Subsystems PRODUCTION ARCHITECTURE <a href="#">[Demo ↗]</a></div> <div><div><div>Procedural Jigsaw</div><div>Native Image Bridge</div><div>Level Sync OTA</div></div><div><div>Offline Leaderboard</div><div>IAP Currency State</div><div>Argos Localization</div></div></div> | <div>Gameplay Prototypes R&amp;D &amp; SOURCE REPOS <a href="#">[Demos ↗]</a></div> <div><div><div><a href="#">This Is Blast ↗</a></div><div><a href="#">Arrow Strike ↗</a></div><div><a href="#">Tank Battle 3D ↗</a></div></div><div><div><a href="#">Bullet Chess ↗</a></div><div><a href="#">Tycoon RTS ↗</a></div><div><a href="#">Trivologist ↗</a></div></div></div> |

PROFESSIONAL STUDIO EXPERIENCE

Vasundhara Game Studio | AI Product Developer (Specialization in Unity) Surat, GJ • Dec 2025 – Current

- Procedural Jigsaw Engine (Calm Jigsaw):** Architected algorithmic N×N puzzle generator using 9 mathematical shape signatures, eliminating static art dependencies with dynamic UV quad generation.
- Native Image Bridge & OTA Sync:** Engineered C# bridge over native Android/iOS APIs for remote image caching, 3-tier downsampling, and non-blocking OTA manifest level delivery.
- Deterministic Logic (MeowSweeper):** Implemented backtracking Sudoku solver state machine, interactive cat mechanics, and persistent powerup transaction wallet.
- Zero-Cost Offline Leaderboard:** Engineered on-device simulated competitive seasons with bot population generation, dynamic rank tiers, and idempotent reward verification with zero backend costs.
- Additional Systems & Tooling:** Developed hybrid mechanics (Calm Solitaire), Apple/Google IAP transaction state, and automated Argos IPC tool for offline multi-language localization.

Appzia | Unity3D Systems & Gameplay Developer Pune, India • June 2024 – July 2025

- Stockfish AI & Real-Time Chess (Bullet Chess):** Architected fast-paced multiplayer chess engine integrated with Stockfish API for live move evaluation, client-side board validation, low-latency clock synchronization, and dynamic tournament brackets.
- Arcade Simulation & Economy Systems (Pinnacle):** Led development of Pinnacle gaming simulation replicating physical machines, implementing deterministic state machines, secure Apple/Google In-App Purchases (IAP), and memory pooling locked at 60 FPS.

StudioKrew | Unity Game Developer (Remote) Delhi, India (Remote) • March 2024 – May 2024

- Solo Card Game Architecture (FreeCell & Solitaire):** Delivered complete end-to-end development independently, architecting deterministic card stack rules, memento-based undo/redo state history, and responsive touch controls.
- Dice Physics & Scoring State Machine (Yahtzee):** Engineered probabilistic dice-rolling physics, category score validation state machines, and turn-based round flow with fluid particle animations.
- Grid Puzzle Engine & Serialization (BrokenTiles):** Developed reactive grid-matching algorithms with cascading tile clearing, implementing lightweight JSON save serialization for instant session recovery.

SHIPPED COMMERCIAL TITLES (GOOGLE PLAY STORE)

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| <b>MeowSweeper</b> <a href="#">[Play Store ↗]</a><br>Sudoku Brain Puzzle<br>Sudoku cat logic puzzle with deterministic solver, simulated offline bot leaderboards, and persistent powerup wallet. | <b>Calm Jigsaw</b> <a href="#">[Play Store ↗]</a><br>Procedural HD Puzzles<br>Procedural NxN jigsaw puzzle engine utilizing 9 mathematical shape signatures with non-blocking OTA level sync. | <b>Calm Solitaire</b> <a href="#">[Play Store ↗]</a><br>Jigsaw & Card Hybrid<br>Hybrid jigsaw fitting and solitaire card stacking game with custom UI shaders and zero runtime asset overhead. |
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CORE ARCHITECTURAL SYSTEMS BUILT (NDA-COMPLIANT BLUEPRINTS)

- **Procedural Jigsaw Board Renderer (Core Gameplay Logic):** Engineered a standalone NxN jigsaw piece generator where 9 mathematical tab/blank shape signatures cover any arbitrary grid size without per-N art assets. Includes baked per-piece UV quads, tab overhang masks, and responsive phone vs. tablet viewport scaling.
- **Native Multi-Tier Image Pipeline (Platform Services · Android & iOS):** Built a unified C# surface over native Android & iOS bridges that fetches, resizes, and caches remote images with bounded on-disk eviction, download queue de-duplication, resolution tiers (256/512/1024), and an auto-resuming offline connectivity monitor.
- **Non-Blocking Level Sync (Platform Services · Background OTA):** Implemented a background manifest hash-diffing content synchronizer that downloads new level packs over-the-air without hitching frame budget or blocking startup, safely staging files to apply on next cold boot.
- **Zero-Cost Offline Leaderboard & Bot Simulation (LiveOps & Telemetry):** Simulated live competitive seasons and ranking tiers entirely on-device with zero server costs, featuring a bot population generator, user-centered windowed rank calculation, season resets, and idempotent reward claim verification.
- **Argos Offline Localization Tool (Editor Tooling · IPC):** Designed a Unity Editor utility crossing the OS process boundary using JSON over stdin/stdout to invoke an offline Python machine translation engine (argotranslate), updating Unity Localization tables automatically without external API costs.

FEATURED GAMEPLAY REPOSITORIES & R&D LAB (GITLAB / SOURCE ARCHITECTURE)

- **Bullet Chess Engine (Stockfish AI & Netcode):** [\[GitLab ↗\]](#) — Fast-paced multiplayer chess engine integrated with Stockfish API for real-time move evaluation, client-side board validation, low-latency clock synchronization, and dynamic tournament brackets.
- **Arrow Strike (Spatial Maze Escape & Graph Logic):** [\[GitLab ↗\]](#) — Procedural infinite level puzzle engine implementing spatial graph search algorithms, directional arrow collision resolution, and zero-allocation runtime grid manipulation.
- **This Is Blast (Stack-Based Matching Queue Engine):** [\[GitLab ↗\]](#) — Stack-based inventory matching mechanics with custom FIFO/LIFO queue data structures, cascading block jam resolution, and animated unit property clearing.
- **Tycoon RTS & Autonomous Agent Dispatch:** [\[GitLab ↗\]](#) — 3D isometric grid placement simulation featuring deterministic economic loops, resource harvesting cycles, and autonomous AI worker agent state machines.

PRODUCTION ENGINEERING & PERFORMANCE STANDARDS

- **Zero-Allocation Runtime & GC Elimination:** Enforce zero per-frame managed heap allocations in core game loops (Update/FixedUpdate), utilizing custom generic object pooling, non-allocating LINQ alternatives, and value-type struct caches to eliminate garbage collection hitching.
- **60 FPS Mobile Performance Budgeting:** Conduct profiling-driven development using Unity Profiler, Memory Profiler, and RenderDoc; enforce strict mobile budgets via SRP Batcher, GPU instancing, texture atlasing (ASTC compression), and dynamic draw call culling.
- **Data-Driven Architecture & Decoupled State:** Architect modular gameplay subsystems with ScriptableObjects for configuration and game events, implementing Model-View-Presenter (MVP) patterns and lightweight JSON serialization for seamless offline state recovery.
- **AI-Augmented Development Pipeline:** Leverage state-of-the-art AI tooling (Claude 3.5/3.7, ChatGPT, OpenAI Codex) to accelerate architectural reviews, verify complex mathematical puzzles, prototype HLSL shaders, and automate Unity Editor utilities.

TECHNICAL SKILLS & TALENT TREE

- **Engines & Languages:** Unity3D (2020 – 2023 / Unity 6), C#, Python, HLSL ShaderLab
- **Networking & Netcode:** Real-Time Multiplayer Netcode, Client-Side Prediction, Deterministic State Sync, REST APIs, JSON
- **AI & Algorithmics:** Stockfish API, Unity ML-Agents, A\* Pathfinding, Procedural Map & Mesh Generation
- **Monetization & LiveOps:** Unity In-App Purchase (IAP for Apple/Google), Local JSON Caching, Ads Mediation SDKs
- **Art & Workflow Tools:** Blender (3D Modeling, UV Unwrapping, Retopology, LODs), Figma, Android Studio
- **DevOps & Testing:** Git, GitLab, Unity Test Framework, Android Device Profiler, ADB